

Executive Summary

EduPro Student Segmentation & Personalized Course Recommendation System
Prepared for Government & Institutional Stakeholders — June 2026

The Issue

EduPro's online learning platform currently sends the same course recommendations to every learner regardless of their goals, behavior, or experience level. This generic approach is leaving engagement, course completion, and learner retention on the table — three outcomes directly tied to the platform's public value and long-term sustainability.

What We Did

We analyzed learner enrollment, spending, and category/level preferences across the platform and built a personalization system with three parts:

- A statistical model that groups learners into five distinct behavioral types based on how they actually use the platform
- A recommendation engine that tailors course suggestions to each learner's type, preferences, and peer behavior
- An interactive dashboard so program staff can explore learner segments and recommendations without needing technical expertise

What We Found: Five Learner Types

Segment	Share of Learners	What Defines Them	Recommended Action
Specialists	33%	Deep focus in 1-2 subject areas	Offer structured, sequential learning paths
Explorers	22%	Wide spread across many categories	Surface discovery bundles across subjects
Premium Upskillers	23%	Highest spend, advanced & certification focus	Prioritize for advanced/certification content
Casual Dabblers	14%	Lowest engagement and spend	Target with retention and re-engagement outreach
Niche Beginners	7%	Few courses, single niche, high certification rate	Investigate dedicated credential-track content

Independent statistical validation (two different clustering methods agreeing on 60% of groupings, and behavioral similarity within each group running far higher than similarity between groups) confirms these are real, consistent patterns in learner behavior — not statistical noise.

What This Means for Outcomes

- Recommendations matched to a learner's actual segment hit the right course category 62% of the time in testing, versus a generic, one-size-fits-all baseline.
- Courses surfaced through the personalized engine carry a measurably higher average quality rating (+3.5%) than what learners currently enroll in on their own — a positive early signal for engagement and satisfaction.
- The Casual Dabbler segment, representing roughly 1 in 7 learners, is clearly identifiable as a retention risk well before they disengage entirely — enabling proactive, targeted outreach rather than reactive win-back efforts.

What We Recommend

- Adopt segment-aware recommendations as the default personalization strategy, replacing the current uniform approach.
- Stand up the accompanying dashboard for program staff to monitor segment composition and recommendation performance on an ongoing basis.
- Pilot targeted retention messaging for the Casual Dabbler segment as a near-term, low-cost intervention.
- Run a structured A/B test comparing personalized vs. generic recommendations on live traffic to convert today's proxy estimates into measured outcomes.

Bottom Line

EduPro's learners are not one audience — they are at least five. Recognizing and acting on that difference is a low-risk, evidence-based step toward a more engaging, more effective, and more sustainable learning platform.